Filter Summary Report: CG,TIA,Automated,simple,Z2,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z2 Z5: $\frac{Z_2 Z_5 g_m - Z_2 + Z_5}{2 Z_2 g_m + 4}$

$$H(z) = \frac{Z_2 Z_5 g_m - Z_2 + Z_5}{2 Z_2 g_m + 4}$$

2 AP

3 BP

4 BS

5 GE

5.1 GE-1 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5},\ \infty\right)$

$$H(s) = \frac{-C_5 L_5 R_2 R_5 s^2 - R_2 R_5 + s\left(L_5 R_2 R_5 g_m - L_5 R_2 + L_5 R_5\right)}{2 R_2 R_5 g_m + 4 R_5 + s^2\left(2 C_5 L_5 R_2 R_5 g_m + 4 C_5 L_5 R_5\right) + s\left(2 L_5 R_2 g_m + 4 L_5\right)}$$

Parameters:

Q: $\frac{\sqrt{C_5} R_5}{\sqrt{L_5}}$
wo: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$
bandwidth: $\frac{1}{C_5 R_5}$
K-LP: $-\frac{R_2}{2 R_2 g_m + 4}$
K-HP: $-\frac{R_2}{2 R_2 g_m + 4}$
K-BP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
Qz: $-\frac{\sqrt{C_5} R_2 R_5}{\sqrt{L_5} R_2 R_5 g_m - \sqrt{L_5} R_2 + \sqrt{L_5} R_5}$
Wz: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$

5.2 GE-2 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{R_5\left(C_5 L_5 s^2 + 1\right)}{C_5 L_5 s^2 + C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{-C_5 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5 + s^2\left(C_5 L_5 R_2 R_5 g_m - C_5 L_5 R_2 + C_5 L_5 R_5\right)}{2 R_2 g_m + s^2\left(2 C_5 L_5 R_2 g_m + 4 C_5 L_5\right) + s\left(2 C_5 R_2 R_5 g_m + 4 C_5 R_5\right) + 4}$$

Parameters:

Q: $\frac{\sqrt{L_5}}{\sqrt{C_5} R_5}$
wo: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$
bandwidth: $\frac{R_5}{L_5}$
K-LP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
K-HP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
K-BP: $-\frac{R_2}{2 R_2 g_m + 4}$
Qz: $\frac{-\sqrt{L_5} R_2 R_5 g_m + \sqrt{L_5} R_2 - \sqrt{L_5} R_5}{\sqrt{C_5} R_2 R_5}$
Wz: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$

5.3 GE-3 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{C_2 R_5 s + R_5 g_m + s^2 \left(C_2 L_2 R_5 g_m - C_2 L_2\right) - 1}{2 C_2 L_2 g_m s^2 + 4 C_2 s + 2 g_m}$$

Parameters:

Q: $\frac{\sqrt{L_2} g_m}{2\sqrt{C_2}}$
wo: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$
bandwidth: $\frac{2}{L_2 g_m}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: $\frac{R_5 g_m - 1}{2 g_m}$
K-BP: $\frac{R_5}{4}$
Qz: $\frac{\sqrt{L_2} R_5 g_m - \sqrt{L_2}}{\sqrt{C_2} R_5}$
Wz: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$

5.4 GE-4 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s^2 \left(C_2 L_2 R_5 g_m - C_2 L_2\right) + s \left(C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5\right) - 1}{2 C_2 L_2 g_m s^2 + 2 g_m + s \left(2 C_2 R_2 g_m + 4 C_2\right)}$$

Parameters:

Q: $\frac{\sqrt{L_2} g_m}{\sqrt{C_2} R_2 g_m + 2\sqrt{C_2}}$
wo: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$
bandwidth: $\frac{\sqrt{C_2} R_2 g_m + 2\sqrt{C_2}}{\sqrt{C_2} L_2 g_m}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: $\frac{R_5 g_m - 1}{2 g_m}$
K-BP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
Qz: $\frac{\sqrt{L_2} R_5 g_m - \sqrt{L_2}}{\sqrt{C_2} R_2 R_5 g_m - \sqrt{C_2} R_2 + \sqrt{C_2} R_5}$
Wz: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$

5.5 GE-5 $Z(s) = \left(\infty,\ \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5 + s^2 \left(C_2 L_2 R_2 R_5 g_m - C_2 L_2 R_2 + C_2 L_2 R_5\right) + s \left(L_2 R_5 g_m - L_2\right)}{2 L_2 g_m s + 2 R_2 g_m + s^2 \left(2 C_2 L_2 R_2 g_m + 4 C_2 L_2\right) + 4}$$

Parameters:

Q: $\frac{\sqrt{C_2} R_2 g_m + 2\sqrt{C_2}}{\sqrt{L_2} g_m}$
wo: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$
bandwidth: $\frac{g_m}{\sqrt{C_2}\left(\sqrt{C_2} R_2 g_m + 2\sqrt{C_2}\right)}$
K-LP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
K-HP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
K-BP: $\frac{R_5 g_m - 1}{2 g_m}$
Qz: $\frac{\sqrt{C_2} R_2 R_5 g_m - \sqrt{C_2} R_2 + \sqrt{C_2} R_5}{\sqrt{L_2} R_5 g_m - \sqrt{L_2}}$
Wz: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$

5.6 GE-6 $Z(s) = \left(\infty,\ \frac{R_2 \left(C_2 L_2 s^2 + 1\right)}{C_2 L_2 s^2 + C_2 R_2 s + 1},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{C_2 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5 + s^2 \left(C_2 L_2 R_2 R_5 g_m - C_2 L_2 R_2 + C_2 L_2 R_5\right)}{4 C_2 R_2 s + 2 R_2 g_m + s^2 \left(2 C_2 L_2 R_2 g_m + 4 C_2 L_2\right) + 4}$$

Parameters:

Q: $\frac{\sqrt{L_2}R_2g_m+2\sqrt{L_2}}{2\sqrt{C_2}R_2}$
wo: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$
bandwidth: $\frac{2R_2}{\sqrt{L_2}\left(\sqrt{L_2}R_2g_m+2\sqrt{L_2}\right)}$
K-LP: $\frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4}$
K-HP: $\frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4}$
K-BP: $\frac{R_5}{4}$
Qz: $\frac{\sqrt{L_2}R_2R_5g_m-\sqrt{L_2}R_2+\sqrt{L_2}R_5}{\sqrt{C_2}R_2R_5}$
Wz: $\frac{1}{\sqrt{C_2}\sqrt{L_2}}$

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty,\ \frac{1}{C_2s},\ \infty,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{R_5g_m + s\left(C_2R_5 - C_5R_5\right) - 1}{4C_2C_5R_5s^2 + 2g_m + s\left(4C_2 + 2C_5R_5g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}\sqrt{g_m}}{2C_2+C_5R_5g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}}$
bandwidth: $\frac{2C_2+C_5R_5g_m}{2C_2C_5R_5}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_2R_5-C_5R_5}{4C_2+2C_5R_5g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{R_2R_5g_m - R_2 + R_5 + s\left(C_2R_2R_5 - C_5R_2R_5\right)}{4C_2C_5R_2R_5s^2 + 2R_2g_m + s\left(4C_2R_2 + 2C_5R_2R_5g_m + 4C_5R_5\right) + 4}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}\sqrt{R_2g_m+2}}{2C_2R_2+C_5R_2R_5g_m+2C_5R_5}$
wo: $\frac{\sqrt{2}\sqrt{R_2g_m+2}}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}}$
bandwidth: $\frac{2C_2R_2+C_5R_2R_5g_m+2C_5R_5}{2C_2C_5R_2R_5}$
K-LP: $\frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_5-C_5R_2R_5}{4C_2R_2+2C_5R_2R_5g_m+4C_5R_5}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty,\ R_2,\ \infty,\ \infty,\ R_5,\ \infty)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5}{2R_2 g_m + 4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 R_2 s + R_2 g_m + 1}{s\left(2C_5 R_2 g_m + 4C_5\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{-C_5 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5}{2R_2 g_m + s\left(2C_5 R_2 R_5 g_m + 4C_5 R_5\right) + 4}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5\right) + 1}{s\left(2C_5 R_2 g_m + 4C_5\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ L_5 s + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 R_2 s + R_2 g_m + s^2\left(C_5 L_5 R_2 g_m + C_5 L_5\right) + 1}{s\left(2C_5 R_2 g_m + 4C_5\right)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{L_5 s}{C_5 L_5 s^2 + 1},\ \infty\right)$

$$H(s) = \frac{-C_5 L_5 R_2 s^2 - R_2 + s\left(L_5 R_2 g_m + L_5\right)}{2R_2 g_m + s^2\left(2C_5 L_5 R_2 g_m + 4C_5 L_5\right) + 4}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ L_5 s + R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s^2\left(C_5 L_5 R_2 g_m + C_5 L_5\right) + s\left(C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5\right) + 1}{s\left(2C_5 R_2 g_m + 4C_5\right)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1},\ \infty\right)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5 + s^2\left(C_5 L_5 R_2 R_5 g_m - C_5 L_5 R_2 + C_5 L_5 R_5\right) + s\left(L_5 R_2 g_m + L_5\right)}{2R_2 g_m + s^2\left(2C_5 L_5 R_2 g_m + 4C_5 L_5\right) + 4}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{C_2 R_5 s + R_5 g_m - 1}{4C_2 s + 2g_m}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s\left(C_2 - C_5\right)}{4C_2 C_5 s^2 + 2C_5 g_m s}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 R_5 s^2 + g_m + s\left(C_2 + C_5 R_5 g_m - C_5\right)}{4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 L_5 s^3 + C_5 L_5 g_m s^2 + g_m + s\left(C_2 - C_5\right)}{4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5 s}{C_5 L_5 s^2 + 1},\ \infty\right)$

$$H(s) = \frac{L_5 g_m s + s^2\left(C_2 L_5 - C_5 L_5\right) - 1}{4 C_2 C_5 L_5 s^3 + 4 C_2 s + 2 C_5 L_5 g_m s^2 + 2 g_m}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 L_5 s^3 + g_m + s^2\left(C_2 C_5 R_5 + C_5 L_5 g_m\right) + s\left(C_2 + C_5 R_5 g_m - C_5\right)}{4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5},\ \infty\right)$

$$H(s) = \frac{-R_5 + s^2\left(C_2 L_5 R_5 - C_5 L_5 R_5\right) + s\left(L_5 R_5 g_m - L_5\right)}{4 C_2 C_5 L_5 R_5 s^3 + 2 R_5 g_m + s^2\left(4 C_2 L_5 + 2 C_5 L_5 R_5 g_m\right) + s\left(4 C_2 R_5 + 2 L_5 g_m\right)}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 L_5 R_5 s^3 + R_5 g_m + s^2\left(C_2 L_5 + C_5 L_5 R_5 g_m - C_5 L_5\right) + s\left(C_2 R_5 + L_5 g_m\right) - 1}{4 C_2 C_5 L_5 s^3 + 4 C_2 s + 2 C_5 L_5 g_m s^2 + 2 g_m}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5\left(C_5 L_5 s^2 + 1\right)}{C_5 L_5 s^2 + C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 L_5 R_5 s^3 + R_5 g_m + s^2\left(C_5 L_5 R_5 g_m - C_5 L_5\right) + s\left(C_2 R_5 - C_5 R_5\right) - 1}{4 C_2 C_5 L_5 s^3 + 2 g_m + s^2\left(4 C_2 C_5 R_5 + 2 C_5 L_5 g_m\right) + s\left(4 C_2 + 2 C_5 R_5 g_m\right)}$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{C_2 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5}{4 C_2 R_2 s + 2 R_2 g_m + 4}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_2 R_2 - C_5 R_2\right) + 1}{4 C_2 C_5 R_2 s^2 + s\left(2 C_5 R_2 g_m + 4 C_5\right)}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 R_2 R_5 s^2 + R_2 g_m + s\left(C_2 R_2 + C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5\right) + 1}{4 C_2 C_5 R_2 s^2 + s\left(2 C_5 R_2 g_m + 4 C_5\right)}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ L_5s + \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_5R_2s^3 + R_2g_m + s^2(C_5L_5R_2g_m + C_5L_5) + s(C_2R_2 - C_5R_2) + 1}{4C_2C_5R_2s^2 + s(2C_5R_2g_m + 4C_5)}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ \frac{L_5s}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{-R_2 + s^2(C_2L_5R_2 - C_5L_5R_2) + s(L_5R_2g_m + L_5)}{4C_2C_5L_5R_2s^3 + 4C_2R_2s + 2R_2g_m + s^2(2C_5L_5R_2g_m + 4C_5L_5) + 4}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ L_5s + R_5 + \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_5R_2s^3 + R_2g_m + s^2(C_2C_5R_2R_5 + C_5L_5R_2g_m + C_5L_5) + s(C_2R_2 + C_5R_2R_5g_m - C_5R_2 + C_5R_5) + 1}{4C_2C_5R_2s^2 + s(2C_5R_2g_m + 4C_5)}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5},\ \infty\right)$

$$H(s) = \frac{-R_2R_5 + s^2(C_2L_5R_2R_5 - C_5L_5R_2R_5) + s(L_5R_2R_5g_m - L_5R_2 + L_5R_5)}{4C_2C_5L_5R_2R_5s^3 + 2R_2R_5g_m + 4R_5 + s^2(4C_2L_5R_2 + 2C_5L_5R_2R_5g_m + 4C_5L_5R_5) + s(4C_2R_2R_5 + 2L_5R_2g_m + 4L_5)}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_5R_2R_5s^3 + R_2R_5g_m - R_2 + R_5 + s^2(C_2L_5R_2 + C_5L_5R_2R_5g_m - C_5L_5R_2 + C_5L_5R_5) + s(C_2R_2R_5 + L_5R_2g_m + L_5)}{4C_2C_5L_5R_2s^3 + 4C_2R_2s + 2R_2g_m + s^2(2C_5L_5R_2g_m + 4C_5L_5) + 4}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ \frac{R_5\left(C_5L_5s^2+1\right)}{C_5L_5s^2+C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_5R_2R_5s^3 + R_2R_5g_m - R_2 + R_5 + s^2(C_5L_5R_2R_5g_m - C_5L_5R_2 + C_5L_5R_5) + s(C_2R_2R_5 - C_5R_2R_5)}{4C_2C_5L_5R_2s^3 + 2R_2g_m + s^2(4C_2C_5R_2R_5 + 2C_5L_5R_2g_m + 4C_5L_5) + s(4C_2R_2 + 2C_5R_2R_5g_m + 4C_5R_5) + 4}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5g_m + s(C_2R_2R_5g_m - C_2R_2 + C_2R_5) - 1}{2g_m + s(2C_2R_2g_m + 4C_2)}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \infty,\ \infty,\ \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{-C_2C_5R_2s^2 + g_m + s(C_2R_2g_m + C_2 - C_5)}{2C_5g_ms + s^2(2C_2C_5R_2g_m + 4C_2C_5)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{g_m + s^2(C_2C_5R_2R_5g_m - C_2C_5R_2 + C_2C_5R_5) + s(C_2R_2g_m + C_2 + C_5R_5g_m - C_5)}{2C_5g_ms + s^2(2C_2C_5R_2g_m + 4C_2C_5)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^3 \left(C_2 C_5 L_5 R_2 g_m + C_2 C_5 L_5\right) + s^2 \left(-C_2 C_5 R_2 + C_5 L_5 g_m\right) + s \left(C_2 R_2 g_m + C_2 - C_5\right)}{2 C_5 g_m s + s^2 \left(2 C_2 C_5 R_2 g_m + 4 C_2 C_5\right)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5 s}{C_5 L_5 s^2 + 1},\ \infty\right)$

$$H(s) = \frac{-C_2 C_5 L_5 R_2 s^3 + s^2 \left(C_2 L_5 R_2 g_m + C_2 L_5 - C_5 L_5\right) + s \left(-C_2 R_2 + L_5 g_m\right) - 1}{2 C_5 L_5 g_m s^2 + 2 g_m + s^3 \left(2 C_2 C_5 L_5 R_2 g_m + 4 C_2 C_5 L_5\right) + s \left(2 C_2 R_2 g_m + 4 C_2\right)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^3 \left(C_2 C_5 L_5 R_2 g_m + C_2 C_5 L_5\right) + s^2 \left(C_2 C_5 R_2 R_5 g_m - C_2 C_5 R_2 + C_2 C_5 R_5 + C_5 L_5 g_m\right) + s \left(C_2 R_2 g_m + C_2 + C_5 R_5 g_m - C_5\right)}{2 C_5 g_m s + s^2 \left(2 C_2 C_5 R_2 g_m + 4 C_2 C_5\right)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5},\ \infty\right)$

$$H(s) = \frac{-C_2 C_5 L_5 R_2 R_5 s^3 - R_5 + s^2 \left(C_2 L_5 R_2 R_5 g_m - C_2 L_5 R_2 + C_2 L_5 R_5 - C_5 L_5 R_5\right) + s \left(-C_2 R_2 R_5 + L_5 R_5 g_m - L_5\right)}{2 R_5 g_m + s^3 \left(2 C_2 C_5 L_5 R_2 R_5 g_m + 4 C_2 C_5 L_5 R_5\right) + s^2 \left(2 C_2 L_5 R_2 g_m + 4 C_2 L_5 + 2 C_5 L_5 R_5 g_m\right) + s \left(2 C_2 R_2 R_5 g_m + 4 C_2 R_5 + 2 L_5 g_m\right)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1},\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s^3 \left(C_2 C_5 L_5 R_2 R_5 g_m - C_2 C_5 L_5 R_2 + C_2 C_5 L_5 R_5\right) + s^2 \left(C_2 L_5 R_2 g_m + C_2 L_5 + C_5 L_5 R_5 g_m - C_5 L_5\right) + s \left(C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5 + L_5 g_m\right) - 1}{2 C_5 L_5 g_m s^2 + 2 g_m + s^3 \left(2 C_2 C_5 L_5 R_2 g_m + 4 C_2 C_5 L_5\right) + s \left(2 C_2 R_2 g_m + 4 C_2\right)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5 \left(C_5 L_5 s^2 + 1\right)}{C_5 L_5 s^2 + C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s^3 \left(C_2 C_5 L_5 R_2 R_5 g_m - C_2 C_5 L_5 R_2 + C_2 C_5 L_5 R_5\right) + s^2 \left(-C_2 C_5 R_2 R_5 + C_5 L_5 R_5 g_m - C_5 L_5\right) + s \left(C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5 - C_5 R_5\right) - 1}{2 g_m + s^3 \left(2 C_2 C_5 L_5 R_2 g_m + 4 C_2 C_5 L_5\right) + s^2 \left(2 C_2 C_5 R_2 R_5 g_m + 4 C_2 C_5 R_5 + 2 C_5 L_5 g_m\right) + s \left(2 C_2 R_2 g_m + 4 C_2 + 2 C_5 R_5 g_m\right)}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_2 C_5 L_2 s^3 + C_2 L_2 g_m s^2 + g_m + s \left(C_2 - C_5\right)}{2 C_2 C_5 L_2 g_m s^3 + 4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{-C_2 C_5 L_2 R_5 s^3 + R_5 g_m + s^2 \left(C_2 L_2 R_5 g_m - C_2 L_2\right) + s \left(C_2 R_5 - C_5 R_5\right) - 1}{2 C_2 C_5 L_2 R_5 g_m s^3 + 2 g_m + s^2 \left(4 C_2 C_5 R_5 + 2 C_2 L_2 g_m\right) + s \left(4 C_2 + 2 C_5 R_5 g_m\right)}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^3 \left(C_2 C_5 L_2 R_5 g_m - C_2 C_5 L_2\right) + s^2 \left(C_2 C_5 R_5 + C_2 L_2 g_m\right) + s \left(C_2 + C_5 R_5 g_m - C_5\right)}{2 C_2 C_5 L_2 g_m s^3 + 4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_2L_5g_m s^4 + g_m + s^3\left(-C_2C_5L_2 + C_2C_5L_5\right) + s^2\left(C_2L_2g_m + C_5L_5g_m\right) + s\left(C_2 - C_5\right)}{2C_2C_5L_2g_m s^3 + 4C_2C_5 s^2 + 2C_5g_m s}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5 s}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5 s^4 + C_2L_2L_5g_m s^3 + L_5g_m s + s^2\left(-C_2L_2 + C_2L_5 - C_5L_5\right) - 1}{2C_2C_5L_2L_5g_m s^4 + 4C_2C_5L_5 s^3 + 4C_2 s + 2g_m + s^2\left(2C_2L_2g_m + 2C_5L_5g_m\right)}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_2L_5g_m s^4 + g_m + s^3\left(C_2C_5L_2R_5g_m - C_2C_5L_2 + C_2C_5L_5\right) + s^2\left(C_2C_5R_5 + C_2L_2g_m + C_5L_5g_m\right) + s\left(C_2 + C_5R_5g_m - C_5\right)}{2C_2C_5L_2g_m s^3 + 4C_2C_5 s^2 + 2C_5g_m s}$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5R_5 s}{C_5L_5R_5s^2+L_5s+R_5},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5R_5 s^4 - R_5 + s^3\left(C_2L_2L_5R_5g_m - C_2L_2L_5\right) + s^2\left(-C_2L_2R_5 + C_2L_5R_5 - C_5L_5R_5\right) + s\left(L_5R_5g_m - L_5\right)}{2C_2C_5L_2L_5R_5g_m s^4 + 2R_5g_m + s^3\left(4C_2C_5L_5R_5 + 2C_2L_2L_5g_m\right) + s^2\left(2C_2L_2R_5g_m + 4C_2L_5 + 2C_5L_5R_5g_m\right) + s\left(4C_2R_5 + 2L_5g_m\right)}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{R_5g_m + s^4\left(C_2C_5L_2L_5R_5g_m - C_2C_5L_2L_5\right) + s^3\left(C_2C_5L_5R_5 + C_2L_2L_5g_m\right) + s^2\left(C_2L_2R_5g_m - C_2L_2 + C_2L_5 + C_5L_5R_5g_m - C_5L_5\right) + s\left(C_2R_5 + L_5g_m\right) - 1}{2C_2C_5L_2L_5g_m s^4 + 4C_2C_5L_5 s^3 + 4C_2 s + 2g_m + s^2\left(2C_2L_2g_m + 2C_5L_5g_m\right)}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5\left(C_5L_5s^2+1\right)}{C_5L_5s^2+C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{R_5g_m + s^4\left(C_2C_5L_2L_5R_5g_m - C_2C_5L_2L_5\right) + s^3\left(-C_2C_5L_2R_5 + C_2C_5L_5R_5\right) + s^2\left(C_2L_2R_5g_m - C_2L_2 + C_5L_5R_5g_m - C_5L_5\right) + s\left(C_2R_5 - C_5R_5\right) - 1}{2C_2C_5L_2L_5g_m s^4 + 2g_m + s^3\left(2C_2C_5L_2R_5g_m + 4C_2C_5L_5\right) + s^2\left(4C_2C_5R_5 + 2C_2L_2g_m + 2C_5L_5g_m\right) + s\left(4C_2 + 2C_5R_5g_m\right)}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2 s^3 + g_m + s^2\left(-C_2C_5R_2 + C_2L_2g_m\right) + s\left(C_2R_2g_m + C_2 - C_5\right)}{2C_2C_5L_2g_m s^3 + 2C_5g_m s + s^2\left(2C_2C_5R_2g_m + 4C_2C_5\right)}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2R_5 s^3 + R_5g_m + s^2\left(-C_2C_5R_2R_5 + C_2L_2R_5g_m - C_2L_2\right) + s\left(C_2R_2R_5g_m - C_2R_2 + C_2R_5 - C_5R_5\right) - 1}{2C_2C_5L_2R_5g_m s^3 + 2g_m + s^2\left(2C_2C_5R_2R_5g_m + 4C_2C_5R_5 + 2C_2L_2g_m\right) + s\left(2C_2R_2g_m + 4C_2 + 2C_5R_5g_m\right)}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^3\left(C_2C_5L_2R_5g_m - C_2C_5L_2\right) + s^2\left(C_2C_5R_2R_5g_m - C_2C_5R_2 + C_2C_5R_5 + C_2L_2g_m\right) + s\left(C_2R_2g_m + C_2 + C_5R_5g_m - C_5\right)}{2C_2C_5L_2g_m s^3 + 2C_5g_m s + s^2\left(2C_2C_5R_2g_m + 4C_2C_5\right)}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_2L_5g_m s^4 + g_m + s^3\left(-C_2C_5L_2 + C_2C_5L_5R_2g_m + C_2C_5L_5\right) + s^2\left(-C_2C_5R_2 + C_2L_2g_m + C_5L_5g_m\right) + s\left(C_2R_2g_m + C_2 - C_5\right)}{2C_2C_5L_2g_m s^3 + 2C_5g_m s + s^2\left(2C_2C_5R_2g_m + 4C_2C_5\right)}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5 s}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5s^4 + s^3\left(-C_2C_5L_5R_2 + C_2L_2L_5g_m\right) + s^2\left(-C_2L_2 + C_2L_5R_2g_m + C_2L_5 - C_5L_5\right) + s\left(-C_2R_2 + L_5g_m\right) - 1}{2C_2C_5L_2L_5g_m s^4 + 2g_m + s^3\left(2C_2C_5L_5R_2g_m + 4C_2C_5L_5\right) + s^2\left(2C_2L_2g_m + 2C_5L_5g_m\right) + s\left(2C_2R_2g_m + 4C_2\right)}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ L_5 s + R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2C_5L_2L_5g_m s^4 + g_m + s^3\left(C_2C_5L_2R_5g_m - C_2C_5L_2 + C_2C_5L_5R_2g_m + C_2C_5L_5\right) + s^2\left(C_2C_5R_2R_5g_m - C_2C_5R_2 + C_2C_5R_5 + C_2L_2g_m + C_5L_5g_m\right) + s\left(C_2R_2g_m + C_2 + C_5R_5g_m - C_5\right)}{2C_2C_5L_2g_m s^3 + 2C_5g_m s + s^2\left(2C_2C_5R_2g_m + 4C_2C_5\right)}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{L_5R_5 s}{C_5L_5R_5s^2+L_5s+R_5},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5R_5s^4 - R_5 + s^3\left(-C_2C_5L_5R_2R_5 + C_2L_2L_5R_5g_m - C_2L_2L_5\right) + s^2\left(-C_2L_2R_5 + C_2L_5R_2R_5g_m - C_2L_5R_2 + C_2L_5R_5 - C_5L_5R_5\right) + s\left(-C_2R_2R_5 + L_5R_5g_m - L_5\right)}{2C_2C_5L_2L_5R_5g_m s^4 + 2R_5g_m + s^3\left(2C_2C_5L_5R_2R_5g_m + 4C_2C_5L_5R_5 + 2C_2L_2L_5g_m\right) + s^2\left(2C_2L_2R_5g_m + 2C_2L_5R_2g_m + 4C_2L_5 + 2C_5L_5R_5g_m\right) + s\left(2C_2R_2R_5g_m + 4C_2R_5 + 2L_5g_m\right)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{R_5g_m + s^4\left(C_2C_5L_2L_5R_5g_m - C_2C_5L_2L_5\right) + s^3\left(C_2C_5L_5R_2R_5g_m - C_2C_5L_5R_2 + C_2C_5L_5R_5 + C_2L_2L_5g_m\right) + s^2\left(C_2L_2R_5g_m - C_2L_2 + C_2L_5R_2g_m + C_2L_5 + C_5L_5R_5g_m - C_5L_5\right) + s\left(C_2R_2R_5g_m - C_2R_2 + C_2R_5 + L_5g_m\right) - 1}{2C_2C_5L_2L_5g_m s^4 + 2g_m + s^3\left(2C_2C_5L_5R_2g_m + 4C_2C_5L_5\right) + s^2\left(2C_2L_2g_m + 2C_5L_5g_m\right) + s\left(2C_2R_2g_m + 4C_2\right)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5\left(C_5L_5s^2+1\right)}{C_5L_5s^2+C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{R_5g_m + s^4\left(C_2C_5L_2L_5R_5g_m - C_2C_5L_2L_5\right) + s^3\left(-C_2C_5L_2R_5 + C_2C_5L_5R_2R_5g_m - C_2C_5L_5R_2 + C_2C_5L_5R_5\right) + s^2\left(-C_2C_5R_2R_5 + C_2L_2R_5g_m - C_2L_2 + C_5L_5R_5g_m - C_5L_5\right) + s\left(C_2R_2R_5g_m - C_2R_2 + C_2R_5 - C_5R_5\right) - 1}{2C_2C_5L_2L_5g_m s^4 + 2g_m + s^3\left(2C_2C_5L_2R_5g_m + 2C_2C_5L_5R_2g_m + 4C_2C_5L_5\right) + s^2\left(2C_2C_5R_2R_5g_m + 4C_2C_5R_5 + 2C_2L_2g_m + 2C_5L_5g_m\right) + s\left(2C_2R_2g_m + 4C_2 + 2C_5R_5g_m\right)}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2R_2s^3 + R_2g_m + s^2\left(C_2L_2R_2g_m + C_2L_2 - C_5L_2\right) + s\left(-C_5R_2 + L_2g_m\right) + 1}{2C_5L_2g_m s^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2R_2R_5s^3 + R_2R_5g_m - R_2 + R_5 + s^2\left(C_2L_2R_2R_5g_m - C_2L_2R_2 + C_2L_2R_5 - C_5L_2R_5\right) + s\left(-C_5R_2R_5 + L_2R_5g_m - L_2\right)}{2R_2g_m + s^3\left(2C_2C_5L_2R_2R_5g_m + 4C_2C_5L_2R_5\right) + s^2\left(2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_2R_5g_m\right) + s\left(2C_5R_2R_5g_m + 4C_5R_5 + 2L_2g_m\right) + 4}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_2g_m + s^3\left(C_2C_5L_2R_2R_5g_m - C_2C_5L_2R_2 + C_2C_5L_2R_5\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_5L_2R_5g_m - C_5L_2\right) + s\left(C_5R_2R_5g_m - C_5R_2 + C_5R_5 + L_2g_m\right) + 1}{2C_5L_2g_m s^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.57 X-INVALID-ORDER-57 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ L_5s+\frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{R_2g_m + s^4\left(C_2C_5L_2L_5R_2g_m + C_2C_5L_2L_5\right) + s^3\left(-C_2C_5L_2R_2 + C_5L_2L_5g_m\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 - C_5L_2 + C_5L_5R_2g_m + C_5L_5\right) + s\left(-C_5R_2 + L_2g_m\right) + 1}{2C_5L_2g_ms^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.58 X-INVALID-ORDER-58 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \frac{L_5s}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5R_2s^4 - R_2 + s^3\left(C_2L_2L_5R_2g_m + C_2L_2L_5 - C_5L_2L_5\right) + s^2\left(-C_2L_2R_2 - C_5L_5R_2 + L_2L_5g_m\right) + s\left(-L_2 + L_5R_2g_m + L_5\right)}{2C_5L_2L_5g_ms^3 + 2L_2g_ms + 2R_2g_m + s^4\left(2C_2C_5L_2L_5R_2g_m + 4C_2C_5L_2L_5\right) + s^2\left(2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_5R_2g_m + 4C_5L_5\right) + 4}$$

9.59 X-INVALID-ORDER-59 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ L_5s+R_5+\frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{R_2g_m + s^4\left(C_2C_5L_2L_5R_2g_m + C_2C_5L_2L_5\right) + s^3\left(C_2C_5L_2R_2R_5g_m - C_2C_5L_2R_2 + C_2C_5L_2R_5 + C_5L_2L_5g_m\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_5L_2R_5g_m - C_5L_2 + C_5L_5R_2g_m + C_5L_5\right) + s\left(C_5R_2R_5g_m - C_5R_2 + C_5R_5 + L_2g_m\right) + 1}{2C_5L_2g_ms^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.60 X-INVALID-ORDER-60 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5R_2R_5s^4 - R_2R_5 + s^3\left(C_2L_2L_5R_2R_5g_m - C_2L_2L_5R_2 + C_2L_2L_5R_5 - C_5L_2L_5R_5\right) + s^2\left(-C_2L_2R_2R_5 - C_5L_5R_2R_5 + L_2L_5R_5g_m - L_2L_5\right) + s\left(-L_2R_5 + L_5R_2R_5g_m - L_5R_2 + L_5R_5\right)}{2R_2R_5g_m + 4R_5 + s^4\left(2C_2C_5L_2L_5R_2R_5g_m + 4C_2C_5L_2L_5R_5\right) + s^3\left(2C_2L_2L_5R_2g_m + 4C_2L_2L_5 + 2C_5L_2L_5R_5g_m\right) + s^2\left(2C_2L_2R_2R_5g_m + 4C_2L_2R_5 + 2C_5L_5R_2R_5g_m + 4C_5L_5R_5 + 2L_2L_5g_m\right) + s\left(2L_2R_5g_m + 2L_5R_2g_m + 4L_5\right)}$$

9.61 X-INVALID-ORDER-61 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{R_2R_5g_m - R_2 + R_5 + s^4\left(C_2C_5L_2L_5R_2R_5g_m - C_2C_5L_2L_5R_2 + C_2C_5L_2L_5R_5\right) + s^3\left(C_2L_2L_5R_2g_m + C_2L_2L_5 + C_5L_2L_5R_5g_m - C_5L_2L_5\right) + s^2\left(C_2L_2R_2R_5g_m - C_2L_2R_2 + C_2L_2R_5 + C_5L_5R_2R_5g_m - C_5L_5R_2 + C_5L_5R_5 + L_2L_5g_m\right) + s\left(L_2R_5g_m - L_2 + L_5R_2g_m\right.}{2C_5L_2L_5g_ms^3 + 2L_2g_ms + 2R_2g_m + s^4\left(2C_2C_5L_2L_5R_2g_m + 4C_2C_5L_2L_5\right) + s^2\left(2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_5R_2g_m + 4C_5L_5\right) + 4}$$

9.62 X-INVALID-ORDER-62 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \frac{R_5\left(C_5L_5s^2+1\right)}{C_5L_5s^2+C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{R_2R_5g_m - R_2 + R_5 + s^4\left(C_2C_5L_2L_5R_2R_5g_m - C_2C_5L_2L_5R_2 + C_2C_5L_2L_5R_5\right) + s^3\left(-C_2C_5L_2R_2R_5 + C_5L_2L_5R_5g_m - C_5L_2L_5\right) + s^2\left(C_2L_2R_2R_5g_m - C_2L_2R_2 + C_2L_2R_5 - C_5L_2R_5 + C_5L_5R_2R_5g_m - C_5L_5R_2 + C_5L_5R_5\right) + s\left(-C_5R_2R_5 + L_2R_5g_m - L_2\right)}{2R_2g_m + s^4\left(2C_2C_5L_2L_5R_2g_m + 4C_2C_5L_2L_5\right) + s^3\left(2C_2C_5L_2R_2R_5g_m + 4C_2C_5L_2R_5 + 2C_5L_2L_5g_m\right) + s^2\left(2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_2R_5g_m + 2C_5L_5R_2g_m + 4C_5L_5\right) + s\left(2C_5R_2R_5g_m + 4C_5R_5 + 2L_2g_m\right) + 4}$$

9.63 X-INVALID-ORDER-63 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2R_2s^3 + R_2g_m + s^2\left(C_2L_2R_2g_m + C_2L_2\right) + s\left(C_2R_2 - C_5R_2\right) + 1}{4C_2C_5R_2s^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.64 X-INVALID-ORDER-64 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2R_2R_5s^3 + R_2R_5g_m - R_2 + R_5 + s^2\left(C_2L_2R_2R_5g_m - C_2L_2R_2 + C_2L_2R_5\right) + s\left(C_2R_2R_5 - C_5R_2R_5\right)}{2R_2g_m + s^3\left(2C_2C_5L_2R_2R_5g_m + 4C_2C_5L_2R_5\right) + s^2\left(4C_2C_5R_2R_5 + 2C_2L_2R_2g_m + 4C_2L_2\right) + s\left(4C_2R_2 + 2C_5R_2R_5g_m + 4C_5R_5\right) + 4}$$

9.65 X-INVALID-ORDER-65 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ R_5+\frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{R_2g_m + s^3\left(C_2C_5L_2R_2R_5g_m - C_2C_5L_2R_2 + C_2C_5L_2R_5\right) + s^2\left(C_2C_5R_2R_5 + C_2L_2R_2g_m + C_2L_2\right) + s\left(C_2R_2 + C_5R_2R_5g_m - C_5R_2 + C_5R_5\right) + 1}{4C_2C_5R_2s^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.66 X-INVALID-ORDER-66 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ L_5s+\frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{R_2g_m + s^4\left(C_2C_5L_2L_5R_2g_m + C_2C_5L_2L_5\right) + s^3\left(-C_2C_5L_2R_2 + C_2C_5L_5R_2\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_5L_5R_2g_m + C_5L_5\right) + s\left(C_2R_2 - C_5R_2\right) + 1}{4C_2C_5R_2s^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.67 X-INVALID-ORDER-67 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \frac{L_5s}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5R_2s^4 - R_2 + s^3\left(C_2L_2L_5R_2g_m + C_2L_2L_5\right) + s^2\left(-C_2L_2R_2 + C_2L_5R_2 - C_5L_5R_2\right) + s\left(L_5R_2g_m + L_5\right)}{4C_2C_5L_5R_2s^3 + 4C_2R_2s + 2R_2g_m + s^4\left(2C_2C_5L_2L_5R_2g_m + 4C_2C_5L_2L_5\right) + s^2\left(2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_5R_2g_m + 4C_5L_5\right) + 4}$$

9.68 X-INVALID-ORDER-68 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ L_5s+R_5+\frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{R_2g_m + s^4\left(C_2C_5L_2L_5R_2g_m + C_2C_5L_2L_5\right) + s^3\left(C_2C_5L_2R_2R_5g_m - C_2C_5L_2R_2 + C_2C_5L_2R_5 + C_2C_5L_5R_2\right) + s^2\left(C_2C_5R_2R_5 + C_2L_2R_2g_m + C_2L_2 + C_5L_5R_2g_m + C_5L_5\right) + s\left(C_2R_2 + C_5R_2R_5g_m - C_5R_2 + C_5R_5\right) + 1}{4C_2C_5R_2s^2 + s^3\left(2C_2C_5L_2R_2g_m + 4C_2C_5L_2\right) + s\left(2C_5R_2g_m + 4C_5\right)}$$

9.69 X-INVALID-ORDER-69 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5},\ \infty\right)$

$$H(s) = \frac{-C_2C_5L_2L_5R_2R_5s^4 - R_2R_5 + s^3\left(C_2L_2L_5R_2R_5g_m - C_2L_2L_5R_2 + C_2L_2L_5R_5\right) + s^2\left(-C_2L_2R_2R_5 + C_2L_5R_2R_5 - C_5L_5R_2R_5\right) + s\left(L_5R_2R_5g_m - L_5R_2 + L_5R_5\right)}{2R_2R_5g_m + 4R_5 + s^4\left(2C_2C_5L_2L_5R_2R_5g_m + 4C_2C_5L_2L_5R_5\right) + s^3\left(4C_2C_5L_5R_2R_5 + 2C_2L_2L_5R_2g_m + 4C_2L_2L_5\right) + s^2\left(2C_2L_2R_2R_5g_m + 4C_2L_2R_5 + 4C_2L_5R_2 + 2C_5L_5R_2R_5g_m + 4C_5L_5R_5\right) + s\left(4C_2R_2R_5 + 2L_5R_2g_m + 4L_5\right)}$$

9.70 X-INVALID-ORDER-70 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1},\ \infty\right)$

$$H(s) = \frac{R_2R_5g_m - R_2 + R_5 + s^4\left(C_2C_5L_2L_5R_2R_5g_m - C_2C_5L_2L_5R_2 + C_2C_5L_2L_5R_5\right) + s^3\left(C_2C_5L_5R_2R_5 + C_2L_2L_5R_2g_m + C_2L_2L_5\right) + s^2\left(C_2L_2R_2R_5g_m - C_2L_2R_2 + C_2L_2R_5 + C_2L_5R_2 + C_5L_5R_2R_5g_m - C_5L_5R_2 + C_5L_5R_5\right) + s\left(C_2R_2R_5 + L_5R_2g_m + L_5\right)}{4C_2C_5L_5R_2s^3 + 4C_2R_2s + 2R_2g_m + s^4\left(2C_2C_5L_2L_5R_2g_m + 4C_2C_5L_2L_5\right) + s^2\left(2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_5R_2g_m + 4C_5L_5\right) + 4}$$

9.71 X-INVALID-ORDER-71 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \frac{R_5\left(C_5L_5s^2+1\right)}{C_5L_5s^2+C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{R_2R_5g_m - R_2 + R_5 + s^4\left(C_2C_5L_2L_5R_2R_5g_m - C_2C_5L_2L_5R_2 + C_2C_5L_2L_5R_5\right) + s^3\left(-C_2C_5L_2R_2R_5 + C_2C_5L_5R_2R_5\right) + s^2\left(C_2L_2R_2R_5g_m - C_2L_2R_2 + C_2L_2R_5 + C_5L_5R_2R_5g_m - C_5L_5R_2 + C_5L_5R_5\right) + s\left(C_2R_2R_5 - C_5R_2R_5\right)}{2R_2g_m + s^4\left(2C_2C_5L_2L_5R_2g_m + 4C_2C_5L_2L_5\right) + s^3\left(2C_2C_5L_2R_2R_5g_m + 4C_2C_5L_2R_5 + 4C_2C_5L_5R_2\right) + s^2\left(4C_2C_5R_2R_5 + 2C_2L_2R_2g_m + 4C_2L_2 + 2C_5L_5R_2g_m + 4C_5L_5\right) + s\left(4C_2R_2 + 2C_5R_2R_5g_m + 4C_5R_5\right) + 4}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty,\ R_2+\frac{1}{C_2s},\ \infty,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{-C_2C_5R_2R_5s^2 + R_5g_m + s\left(C_2R_2R_5g_m - C_2R_2 + C_2R_5 - C_5R_5\right) - 1}{2g_m + s^2\left(2C_2C_5R_2R_5g_m + 4C_2C_5R_5\right) + s\left(2C_2R_2g_m + 4C_2 + 2C_5R_5g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_5}R_2\sqrt{R_5}g_m\sqrt{\frac{g_m}{R_2g_m+2}}+2\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{R_2g_m+2}}}{C_2R_2g_m+2C_2+C_5R_5g_m}$

wo: $\sqrt{\frac{g_m}{C_2C_5R_2R_5g_m+2C_2C_5R_5}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_2C_5R_2R_5g_m+2C_2C_5R_5}}\left(C_2R_2g_m+2C_2+C_5R_5g_m\right)}{\sqrt{C_2}\sqrt{C_5}R_2\sqrt{R_5}g_m\sqrt{\frac{g_m}{R_2g_m+2}}+2\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{R_2g_m+2}}}$

K-LP: $\frac{R_5g_m-1}{2g_m}$

K-HP: $-\frac{R_2}{2R_2g_m+4}$

K-BP: $\frac{C_2R_2R_5g_m-C_2R_2+C_2R_5-C_5R_5}{2C_2R_2g_m+4C_2+2C_5R_5g_m}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}}$

11 X-PolynomialError